

# NCAIR LMS Onboarding Assistant: A Tool-Routed, Retrieval-Augmented Multilingual Chatbot for Intern Registration Support

Group 5 DSA

**Abstract**— New interns at the National Centre for Artificial Intelligence and Robotics (NCAIR) must complete a ten-step registration on the NCAIR Learning Management System (LMS) by following an eleven-page, image-based manual written in English. This paper presents the NCAIR LMS Onboarding Assistant (version 2.0), a locally hosted, retrieval-augmented chatbot that answers onboarding questions in English, Hausa, Yorùbá, Igbo, and Nigerian Pidgin. A small language model (Llama 3.2 3B served through Ollama) acts as a JSON-schema-constrained router that detects the user’s language, rewrites the question in English, and selects up to two tools: a portal-link lookup, an onboarding-step explainer, and a hybrid BM25-plus-dense knowledge-base search with an abstention signal. A second model call writes a reply grounded only in the returned context, using N-ATLaS for Hausa, Yorùbá, and Igbo, and deterministic guardrails remove invented URLs and enforce required links. On a 76-question benchmark, version 2.0 was compared with the keyword-routed version 1.0: routing accuracy rose from 72.4% to 85.5%, answer completeness from 61.9% to 76.2%, link accuracy from 45.0% to 85.0%, and fact retrieval from 82.8% to 93.1%. The costs were a higher rate of wrongly refusing answerable questions (9.5% to 16.7%) and a longer median response time (2.7 s to 6.0 s, a system-dependent value). The N-ATLaS multilingual pathway, scored on nine Hausa, Yorùbá, and Igbo questions, recorded a pass rate of 55.6%. The results suggest that tool-routed retrieval with explicit grounding rules is a practical foundation for a local onboarding assistant, and that abstention calibration, latency, and multilingual evaluation are the main areas for further work.

**Index Terms**— Retrieval-Augmented Generation (RAG), Onboarding Chatbot, Tool Routing, Hybrid Retrieval, Multilingual NLP, N-ATLaS, Local Language Models.

## I. INTRODUCTION

Onboarding at the National Centre for Artificial Intelligence and Robotics (NCAIR) is a one-time process for interns who are completely new. After orientation, interns walk to the 50-Seater Hall in the e-Government Building, wait their turn in a single line, and finish registering on the NCAIR Learning Management System (LMS) in about 10 to 20 minutes using their own phone [4]. The process depends on supporting documents (an acceptance letter and a National Identification Number), an invitation email, and ten numbered steps, and every step is a place where a first-time user can get stuck: the invitation email lands in spam, a button will not open, a password is rejected, or an account is locked. The official guidance exists as an image-based manual and a programme guideline file, and neither can be queried in the user’s own words or in the user’s own language.

A chatbot can close this gap, but a language model that answers from memory is unreliable for exactly the facts that matter here, such as attendance thresholds, upload limits, and URLs. Retrieval-augmented generation (RAG) [1] mitigates the problem by supplying passages at query time, and unsupported statements in generated text are a well-documented failure that grounding aims to reduce [2]. A single “retrieve then generate” step, however, still has to decide whether a question is about a link, a step in the process, or a policy, and whether the documents contain the

answer at all.

The first version of this assistant, referred to here as version 1.0, used hard-coded keyword rules to choose between three behaviours: returning a link, returning one of four hard-coded step descriptions, or searching a vector index and calling one language model. This paper presents version 2.0, which differs on three fronts. First, routing is done by a language model under a JSON schema generated from a tool registry, so the model can only name tools, pages, and steps that exist. Second, every reply, including link and step replies, is written by the model from retrieved or tool-provided context, with an explicit abstention path when the evidence is weak. Third, language support is built in: the router detects the user’s language and rewrites the question in English for retrieval, and the N-ATLaS model [3] is used to write answers in Hausa, Yorùbá, and Igbo. Deterministic guardrails then check every URL in the reply.

The contributions are: (i) schema-constrained tool routing over a data-driven registry; (ii) hybrid lexical and dense retrieval with a two-level abstention signal; (iii) link guardrails and an extractive offline fallback; and (iv) a 76-question benchmark with a seven-metric head-to-head comparison against version 1.0.

The remainder of this paper is organized as follows. Section II reviews the techniques the system builds on and describes version 1.0. Section III describes the knowledge sources, tool

registry, retrieval, models, benchmark, and deployment. Section IV presents the comparison results and discusses them. Section V walks through the deployed assistant. Section VI concludes the paper, and Section VII outlines directions for future work.

## II. LITERATURE REVIEW

### A. Retrieval-Augmented Generation and Grounding

Lewis et al. [1] introduced retrieval-augmented generation, which pairs a parametric sequence-to-sequence generator with a non-parametric memory reached through a dense retriever, so that generated text can draw on retrieved documents rather than on model weights alone. The survey of Ji et al. [2] documents hallucination in natural language generation, meaning fluent output that is not supported by the source material. This is the failure that matters most for an onboarding assistant, where a wrong deadline, file limit, or link has direct consequences for the user. Both strands motivate the principle used here: the answer model is told to use only the supplied context and to say so when the context does not answer the question.

### B. Tool Use and Routing with Language Models

ReAct [5] showed that language models can interleave reasoning with actions that query external sources, and Toolformer [6] showed that a model can learn which external tools to call and with what arguments. This assistant must run on ordinary local hardware with a 3-billion-parameter model from the Llama 3 family [7], a setting in which unconstrained tool selection is a risk. For that reason the router does not rely on native function calling. It asks the runtime for structured output constrained by a JSON schema, which makes it impossible to name a tool, page, or step that does not exist.

### C. Lexical, Dense, and Hybrid Retrieval

BM25 [8] remains a strong lexical baseline and needs no embedding model. Dense sentence embeddings [9], including multilingual models trained by knowledge distillation [10], match paraphrases that share few words, which matters when a Hausa, Yorùbá, or Igbo question is searched against English passages. Reciprocal rank fusion [11] combines rankings from different retrievers without score calibration. Version 1.0 used a FAISS [12] vector index over sentence embeddings alone. Version 2.0 fuses BM25 with an optional multilingual embedding model and falls back to BM25 when the embedding model is not installed.

### D. Support for Nigerian Languages

N-ATLaS [3] is an eight-billion-parameter model based on Llama 3 for English, Hausa, Yorùbá, and Igbo, released by Awarri Technologies with the Federal Ministry of Communications, Innovation and Digital Economy. Its full weights are about 16 GB, and quantized GGUF builds [13] of roughly 4.9 GB make local use feasible. The model card’s human evaluation, as recorded in the project documentation, rated Yorùbá lowest (2.69 out of 5, against 4.21 for English, 3.98 for Hausa, and 3.87 for Igbo), and Nigerian Pidgin is not a listed language. These properties explain two design decisions: in the default profile, N-ATLaS writes only Hausa, Yorùbá, and Igbo answers, and Pidgin stays with the default model.

Table 1.0. Summary of Reviewed Techniques

| # | Technique                                         | Role in this work                               |
|---|---------------------------------------------------|-------------------------------------------------|
| 1 | Retrieval-augmented generation [1]                | Grounds replies in retrieved passages           |
| 2 | Hallucination analysis [2]                        | Motivates context-only answering and abstention |
| 3 | ReAct [5]                                         | Model-selected actions; basis of the tool loop  |
| 4 | Toolformer [6]                                    | Model decides which tool to call                |
| 5 | BM25 [8]                                          | Lexical half of hybrid retrieval                |
| 6 | Sentence-BERT [9]; multilingual distillation [10] | Dense half; cross-lingual matching              |
| 7 | Reciprocal rank fusion [11]                       | Fuses lexical and dense rankings                |
| 8 | FAISS [12]                                        | Vector index in version 1.0                     |
| 9 | N-ATLaS [3], [13]                                 | Answer model for Hausa, Yorùbá, Igbo            |

### E. Version 1.0 and the Research Gap

Version 1.0 routed with keyword rules: words such as “link” or “open” triggered a link lookup over a hard-coded dictionary, “stuck” or “step” triggered one of four hard-coded step descriptions (the manual describes ten steps), and everything else was searched in a FAISS index (all-MiniLM-L6-v2 embeddings, 500-character chunks with 80 overlap, top two chunks) before a single Llama 3.2 3B call. It had no conversation history, no language handling, and fell back to a support e-mail address whenever the context was insufficient.

The works in Table 1.0 each address one component of the problem. What remains is to combine schema-constrained routing, hybrid retrieval with calibrated abstention, link guardrails, and a specialist Nigerian-language model in one assistant that runs locally, and to measure that combination against a concrete baseline. This paper does so.

### III. METHODOLOGY

#### A. System Overview

The assistant has three cooperating parts: a router, a tool registry with its knowledge sources, and an answer generator followed by guardrails. For a single user turn the pipeline proceeds as follows (Fig. 1):

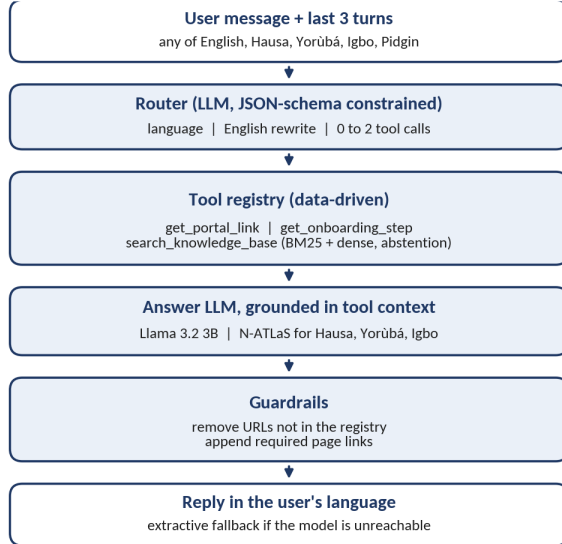


Fig. 1. End-to-end pipeline for a single user turn.

1. Input — the user's message and up to the last three conversation turns are received.
2. Routing — the router model returns one JSON object holding the detected language (English, Hausa, Yorùbá, Igbo, Pidgin, or other), a standalone English rewrite of the question, and zero to two tool calls. The output is constrained by a schema generated from the tool registry. If it cannot be parsed after one retry, or names no valid call, the English question is searched in the knowledge base instead of being keyword-matched.
3. Tool execution — each call runs against the registry (Table 2.0). Tools return context, not final text.
4. Answer generation — a system prompt requires the model to write in the user's language, use only the supplied context, copy URLs exactly, and tell the user when information is missing and to ask an NCAIR staff member at the hall. In the default profile, N-ATLaS is tried first for Hausa, Yorùbá, and Igbo when installed (availability is re-checked every 60 s), and the default model is used if it fails; other model profiles can assign N-ATLaS to answer writing more broadly.
5. Guardrails — URLs in the reply that are not in the page registry or the tool context are removed, and up to two required page links that the model omitted are

appended.

6. Offline degradation — if no model is reachable, the user receives the retrieved text instead of an error.

#### B. Knowledge Sources and Preprocessing

The knowledge base is built from three files. The onboarding manual (11 pages, image-based) [4] was processed with OCR and curated into a structured copy of 23 sections: 10 numbered onboarding steps, 6 troubleshooting sections, 2 rule sections, and 5 information sections. A plain-text programme guideline file adds 6 sections covering location, course curriculum, SIWES-versus-NYSC rules, attendance and grading, troubleshooting, and returning interns. Together they yield 29 retrievable chunks, each capped at 700 characters and split on sentence boundaries.

A page registry holds six portal pages with their URLs, descriptions, and a verification flag. Three of the six (courses, profile, and contact) were carried over from version 1.0, do not appear in the official manual, and are marked unverified. Seven worked examples for the router are kept in a separate file. Because links, steps, prompts, and schemas are all generated from these files, adding a page or a tool is a data edit rather than a code change.

#### C. Tool Registry

Table 2.0. Tool Registry

| Tool                  | Parameter                | Purpose                                                                   |
|-----------------------|--------------------------|---------------------------------------------------------------------------|
| get_portal_link       | page_id (one of 6 pages) | Returns the title, URL, and description of one LMS or NCAIR page          |
| get_onboarding_step   | step (all, or 1 to 10)   | Returns the manual text for a step with its related page link             |
| search_knowledge_base | query (short English)    | Hybrid retrieval; returns passages, related links, and an abstention flag |

#### D. Retrieval and Abstention

Lexical search is a pure-Python BM25 ( $k_1 = 1.5$ ,  $b = 0.75$ ) over tokens that are lower-cased, diacritic-folded (so Yorùbá and Igbo tone marks never block a match), stop-word filtered, and lightly stemmed. When the multilingual embedding model paraphrase-multilingual-MiniLM-L12-v2 [10] is installed, its normalized embeddings are cached on disk, keyed by a hash of the corpus, and fused with BM25 by reciprocal rank fusion with constant 60 [11]. The top four chunks are returned. When the best chunk belongs to a long section, its sibling chunks are added (up to 1,200 characters) so that the sentence that answers the question is not lost to chunking.

The abstention signal, *coverage*, is the idf-weighted share of the query’s terms found in the best passages; terms never seen in the corpus count as maximally informative, so a query such as “WiFi password” scores low. Below 0.25 the answer model is told that no relevant information was found (hard abstain). Between 0.25 and 0.5 the passages are passed on but flagged low-confidence, and the model must say the topic is not covered unless the passages clearly answer the question. The 0.25 threshold was tuned on part of the benchmark as the lowest value with no false abstentions, so benchmark scores are not independent of it.

### E. Models and Generation Settings

Table 3.0. Model, Retrieval, and Decoding Configuration

| Parameter                                 | Value                                                                                                     |
|-------------------------------------------|-----------------------------------------------------------------------------------------------------------|
| Router and answer model (default profile) | llama3.2:3b, served by Ollama [14]                                                                        |
| Multilingual specialist                   | N-ATLaS, quantized GGUF (setup default Q4_K_M, about 4.9 GB); writes Hausa, Yorùbá, and Igbo answers only |
| Router decoding                           | temperature 0.0; up to 200 new tokens; context 4,096                                                      |
| Answer decoding                           | temperature 0.1; repeat penalty 1.12; up to 500 new tokens; context 4,096                                 |
| Tool calls per turn                       | at most 2                                                                                                 |
| Model profiles                            | llama (default); natlas (N-ATLaS for routing and answers); mixed (Llama routes, N-ATLaS answers)          |
| Conversation history                      | last 3 turns                                                                                              |
| Model timeout                             | 180 s                                                                                                     |
| Lexical retrieval                         | BM25, $k_1 = 1.5$ , $b = 0.75$                                                                            |
| Dense retrieval                           | paraphrase-multilingual-MiniLM-L12-v2 (optional)                                                          |
| Fusion                                    | reciprocal rank fusion, constant 60                                                                       |
| Passages returned                         | top 4 plus sibling chunks of the best hit                                                                 |
| Chunk size limit                          | 700 characters                                                                                            |
| Abstention thresholds                     | hard 0.25; soft 0.5                                                                                       |

### F. Evaluation Benchmark and Metrics

The benchmark contains 76 questions written by the project team from the manual and the programme guidelines, not collected from live users. Each case lists the expected tool calls, the facts a correct answer must contain (each a list of acceptable alternatives), the expected links, the gold source sections, and whether the documents answer the question at all. Table 4.0 shows the composition. Of the 76 questions, 64 are in English and 3 each are in Hausa, Yorùbá, Igbo, and Pidgin; 9 should be answered by abstaining, and 2 are follow-ups that depend on history. The multilingual questions are drafts not yet reviewed by native speakers. All 76 are used for evaluation, with no development and test split.

Table 4.0. Composition of the Evaluation Benchmark

| Category            | Cases     |
|---------------------|-----------|
| Knowledge base (kb) | 25        |
| Onboarding steps    | 12        |
| Multilingual        | 12        |
| Portal links        | 10        |
| Out of scope        | 9         |
| Small talk          | 3         |
| Multi-tool          | 3         |
| Follow-up           | 2         |
| <b>Total</b>        | <b>76</b> |

Both versions answered the same 76 questions and were scored with the same helper functions: case-insensitive matching of expected facts and URLs, and phrase-based refusal detection. Because version 1.0 has no history input, follow-up questions were sent to it without history. Its tool choice is inferred from which function it called. Seven metrics are reported (Table 5.0). Answers and grounding use English questions that expect facts and should not be refused, and retrieval uses knowledge-base questions with expected facts. For grounding and speed, lower is better. Speed depends on the system on which it is measured, so it is a changing value, not a constant.

Table 5.0. Evaluation Metrics and Eligible Cases

| Metric    | Definition                                   | Cases |
|-----------|----------------------------------------------|-------|
| Router    | Right tool set chosen                        | 76    |
| Answers   | All expected facts in the reply              | 42    |
| Links     | Expected page URL in the reply               | 20    |
| Retrieval | Expected facts in retrieved passages         | 29    |
| Grounding | Answerable questions wrongly refused         | 42    |
| Speed     | Median seconds per answer                    | 76    |
| N-ATLaS   | N-ATLaS wrote the reply; right tool and link | 9     |

## G. Deployment

The assistant is served as a Gradio [15] chat interface. It is local-only by default, with the public link disabled unless explicitly enabled, and an optional debug trace shows the detected language, router fallback, answer model, tool calls, sources, removed and added URLs, and timings. The language-model layer talks to Ollama over HTTP using only the Python standard library, and a scripted fake client lets the whole pipeline be tested without a model. The attribution required by the N-ATLaS terms is shown in the interface, and the project documentation notes that those terms cap use at 1,000 active users per 30 days without a commercial licence.

## IV. RESULTS

Table 6.0. Seven-Metric Comparison on the 76-Question Benchmark

| Metric                     | Version 1.0 | Version 2.0 | Better when |
|----------------------------|-------------|-------------|-------------|
| Router: right tool         | 72.4%       | 85.5%       | higher      |
| Answers: all facts present | 61.9%       | 76.2%       | higher      |
| Links: right page          | 45.0%       | 85.0%       | higher      |
| Retrieval: facts retrieved | 82.8%       | 93.1%       | higher      |
| Grounding: wrongly refuses | 9.5%        | 16.7%       | lower       |
| Speed: median per answer†  | 2.7 s       | 6.0 s       | lower       |
| N-ATLaS: multilingual pass | -           | 55.6%       | higher      |

† Speed is system-dependent: it varies with hardware, model size and quantization, whether the model is already loaded, and concurrent load. It is therefore a changing value measured on the evaluation system, not a constant. A dash means version 1.0 has no such component.

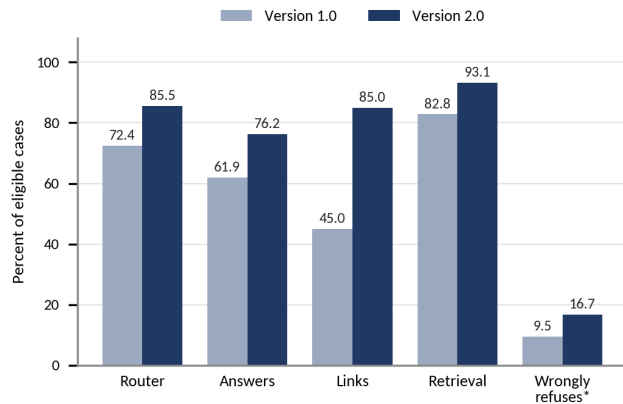


Fig. 2. Versions 1.0 and 2.0 on the five percentage metrics. \*Lower is better for wrongly refused answerable questions.

Table 7.0. Supplementary Retrieval and Speed Measures

| Measure                                      | Version 1.0 | Version 2.0 |
|----------------------------------------------|-------------|-------------|
| Retrieval, English questions only (27 cases) | 88.9%       | 88.9%       |
| Speed, median per answer†                    | 2.7 s       | 6.0 s       |

† System-dependent, as noted under Table 6.0. Version 1.0 searches the raw question with top-2 retrieval; version 2.0 uses the router’s English rewrite with top-4 retrieval, and the aggregate retrieval value in Table 6.0 uses each system’s own input.

Table 8.0. N-ATLaS Pathway Results

| Measure (9 Hausa, Yorùbá, Igbo cases) | Result |
|---------------------------------------|--------|
| Pass rate (N-ATLaS + tool + link)     | 55.6%  |
| Reply written by N-ATLaS              | 9/9    |
| Right tool chosen                     | 7/9    |
| Right link                            | 7/9    |
| Language detected correctly           | 4/9    |

## A. Discussion

**Quality.** Version 2.0 scored higher on all four quality metrics (Table 6.0, Fig. 2). Routing accuracy rose from 72.4% to 85.5% (55 to 65 of 76 questions), complete-fact answers from 61.9% to 76.2% (26 to 32 of 42), correct links from 45.0% to 85.0% (9 to 17 of 20), and fact retrieval from 82.8% to 93.1% (24 to 27 of 29). The link result is the largest change and follows from the design: version 1.0 returned a link only when a keyword rule fired, whereas version 2.0 has the model choose a page from the registry and a guardrail appends the required URL if the reply omits it. On the 27 English knowledge-base questions searched as written, the two versions are level at 88.9% (Table 7.0), so the aggregate retrieval gain comes from how each system forms its search input, which in version 2.0 is the router’s English rewrite, and not from retrieval on identical text.

**Refusals and speed.** Two measures moved in the wrong direction. The share of answerable questions that version 2.0 wrongly refused rose from 9.5% to 16.7% (4 to 7 of 42). Most of version 1.0’s failures were incomplete or incorrect answers rather than refusals, whereas version 2.0 more often says it does not have the information. That is the intended failure mode, but it still costs usefulness. The refusal detector is phrase-based, so a reply that hedges with “I don’t have that information” yet still supplies the fact can be counted as a refusal, which means the figure may overstate the loss of content, although the hedging is itself a usability cost. A plausible contributor is the two-level abstention design, which tells the model to decline unless low-confidence passages clearly answer the question. The benchmark does not isolate this effect, and the thresholds were tuned on the

same questions.

Median time per answer rose from 2.7 s to 6.0 s, about 2.2 times. Every version 2.0 reply needs at least a router call and an answer call, whereas version 1.0's hard-coded link and step branches need no model call. Absolute times depend on the system, including processor or GPU, model size and quantization, whether the model is already resident in memory, and concurrent load, so these values indicate relative cost and should not be read as constants.

**Multilingual pathway.** The N-ATLaS pathway was scored on nine Hausa, Yorùbá, and Igbo questions with a strict criterion: the reply must be written by N-ATLaS, and the router must choose the right tool and the right link (Table 8.0). The pass rate was 55.6% (5 of 9). All nine replies were written by N-ATLaS, so the shortfall comes from routing: the router chose the right tool for 7 of 9 questions and the right link for 7 of 9, and detected the language correctly for 4 of 9. Language detection was therefore the weakest component. Model profiles can assign N-ATLaS to answer writing regardless of the detected language, which allows the written-by count to exceed the detection count. The metric measures whether the specialist model took part in a correctly routed reply, not the linguistic quality of its output, which needs review by native speakers and is not scored here.

**Limitations.** The benchmark was written by the project team from the same documents the assistant retrieves from, it has no held-out split, and the abstention threshold was tuned on part of it, so the scores are in-distribution. The multilingual questions are unreviewed drafts, with only three per language. Fact and link scoring is string matching and cannot judge fluency or faithfulness beyond the presence of required strings. Each version was run once, and one case is worth about 1.3 percentage points on 76 questions, so small differences may not be stable. Three of the six registry links and the support e-mail address are unverified against the live site.

## V. SYSTEM TESTING / APPLICATION WALKTHROUGH

The assistant is a green-themed Gradio chat window titled “NCAIR LMS Assistant” with example prompts in English, Hausa, Yorùbá, and Igbo. A user types a question and the pipeline in Fig. 1 runs. Four interactions illustrate the main paths.

1. *Where do I sign in?* The router selects `get_portal_link` with the sign-in page; the reply contains

<https://lms.ncair.nitda.gov.ng/intern/signin>, and the guardrail appends the link if the model leaves it out.

2. *I'm stuck on step 4.* The router selects `get_onboarding_step` with step 4; the reply is a short numbered list built from the manual text, with button and page names kept as written.
3. *What is the minimum attendance?* The router selects `search_knowledge_base`; the reply is drawn from the retrieved passages, or says the information is not available and points the user to staff at the hall when coverage is too low.
4. *hi there.* No tool is called; the reply is a short greeting with an offer to help with registration, courses, or portal links.

In the debug view, each reply is accompanied by the detected language, tool calls, sources, removed and added URLs, and timings. If the model server is unreachable, the user sees the relevant manual text with a notice instead of an error.

## VI. CONCLUSION

This paper presented the NCAIR LMS Onboarding Assistant (version 2.0), a locally hosted, retrieval-augmented chatbot that uses a schema-constrained language-model router, a data-driven tool registry, hybrid retrieval with two-level abstention, link guardrails, and a specialist model for Hausa, Yorùbá, and Igbo. Against the keyword-routed version 1.0 on a 76-question benchmark, version 2.0 improved routing (72.4% to 85.5%), answer completeness (61.9% to 76.2%), link accuracy (45.0% to 85.0%), and fact retrieval (82.8% to 93.1%). The costs were more wrongly refused answerable questions (9.5% to 16.7%) and a longer median response time (2.7 s to 6.0 s on the evaluation system). The N-ATLaS pathway passed 5 of its nine questions (55.6%), with language detection (4 of 9) the weakest component, so routing for Hausa, Yorùbá, and Igbo needs improvement before its multilingual quality can be claimed. Overall, tool-routed retrieval with explicit grounding rules is a practical foundation for a local onboarding assistant, with abstention calibration, latency, and multilingual evaluation as the main open items.

## VII. FUTURE WORK

Several directions can extend this work. First, the multilingual questions should be reviewed by native speakers and expanded beyond three per language, and the quality of Hausa, Yorùbá, and Igbo replies should be evaluated directly rather than inferred from string matching. Second, a held-out evaluation set is needed so that the abstention thresholds can be re-tuned to reduce wrongly refused questions while still declining out-of-scope ones. Third, latency should be reduced through smaller or accelerated models, caching of repeated questions, and a faster path for simple link and step requests. Fourth, language identification could be made independent of the 3-billion-parameter router so that specialist-model selection is more reliable. Fifth, the unverified registry links and the support e-mail address should be checked against the live LMS and kept in sync with it. Finally, a user study with real interns at onboarding would measure task completion and time to register.

## REFERENCES

- [1] P. Lewis *et al.*, “Retrieval-augmented generation for knowledge-intensive NLP tasks,” in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 33, 2020, pp. 9459–9474.
- [2] Z. Ji *et al.*, “Survey of hallucination in natural language generation,” *ACM Comput. Surv.*, vol. 55, no. 12, Art. 248, 2023.
- [3] NCAIR and Awarri Technologies, “N-ATLaS (NCAIR1/N-ATLaS),” model card, Hugging Face. [Online]. Available: <https://huggingface.co/NCAIR1/N-ATLaS>
- [4] NCAIR, “NCAIR LMS New Student Onboarding Manual,” 11 pp., and “NCAIR LMS & Internship Program Official Guidelines,” internal documents.
- [5] S. Yao *et al.*, “ReAct: Synergizing reasoning and acting in language models,” in *Proc. Int. Conf. Learning Representations (ICLR)*, 2023.
- [6] T. Schick *et al.*, “Toolformer: Language models can teach themselves to use tools,” in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 36, 2023.
- [7] A. Grattafiori *et al.*, “The Llama 3 herd of models,” arXiv preprint arXiv:2407.21783, 2024.
- [8] S. Robertson and H. Zaragoza, “The probabilistic relevance framework: BM25 and beyond,” *Found. Trends Inf. Retr.*, vol. 3, no. 4, pp. 333–389, 2009.
- [9] N. Reimers and I. Gurevych, “Sentence-BERT: Sentence embeddings using Siamese BERT-networks,” in *Proc. EMNLP-IJCNLP*, 2019, pp. 3982–3992.
- [10] N. Reimers and I. Gurevych, “Making monolingual sentence embeddings multilingual using knowledge distillation,” in *Proc. EMNLP*, 2020, pp. 4512–4525.
- [11] G. V. Cormack, C. L. A. Clarke, and S. Büttcher, “Reciprocal rank fusion outperforms Condorcet and individual rank learning methods,” in *Proc. 32nd Int. ACM SIGIR Conf.*, 2009, pp. 758–759.
- [12] J. Johnson, M. Douze, and H. Jégou, “Billion-scale similarity search with GPUs,” arXiv preprint arXiv:1702.08734, 2017.
- [13] “N-ATLaS-GGUF (quantized builds),” Hugging Face repository tosinamuda/N-ATLaS-GGUF. [Online]. Available: <https://huggingface.co/tosinamuda/N-ATLaS-GGUF>
- [14] Ollama, “Ollama: Run large language models locally.” [Online]. Available: <https://ollama.com>
- [15] A. Abid *et al.*, “Gradio: Hassle-free sharing and testing of ML models in the wild,” arXiv preprint arXiv:1906.02569, 2019.